UNIVERSITY OF HAWAIʻI / RESEARCH OPPORTUNITY 03 / GRADUATE RESEARCH

PhD Research Opportunity

Hydrology / Remote sensing / AI for water hazards

From orbit to the streambank: turn observations into better decisions about water.

UNIVERSITY OF HAWAIʻI • DOCTORAL RESEARCH

We're looking for a curious future researcher to connect two big questions: How can satellites help us understand reservoir water levels, and how can sensors, cameras, and AI help communities recognize water-related hazards as they happen?

Research you could shape

- **See reservoirs from space.** Combine satellite imagery and in-water gauges to estimate reservoir levels, track change, and test methods useful for drought resilience and dam safety.
- **Understand hazards on the ground.** Link stream, coastal, and weather observations with camera imagery and historical event records to recognize meaningful flood and hazard conditions.
- **Build and test methods.** Develop reproducible Python/AI/geospatial analyses, validate them against observations, quantify uncertainty, and turn results into papers and usable data products.
- **Work with people who use the data.** Collaborate with researchers, agencies, and communities on what information is actionable and how to communicate it.

What makes a strong fit

- **Academic background.** Bachelor's or master's training in hydrology, geoscience, civil/environmental engineering, geography, remote sensing, computer/data science, or a related field.
- **Quantitative curiosity.** Experience with programming and data analysis (Python preferred), plus interest in geospatial and hydrologic questions.
- **Research mindset.** Eagerness to learn, test assumptions, communicate clearly, and develop an original dissertation with practical value.
- **Great extras.** Satellite imagery, machine learning, computer vision, field sensing, hydrologic modeling, or science communication. You need not know all of these.

Possible work spans both field observations and computational analysis. A thesis direction will be developed with the student rather than treating every project idea as a fixed assignment.

The research directions draw on ongoing project development in environmental hazard awareness and satellite-based reservoir monitoring. Graduate program, campus location, funding package, and start date have not yet been finalized.

Start a conversation

Contact Chris Shuler at cshuler@hawaii.edu with a short note, CV, and your research interests. Formal UH graduate admissions and funding details will be provided once confirmed.

Research opportunity preview, not an official graduate-program admission or funding offer. Project summaries include proposed activities, not guaranteed award commitments.